

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation(BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	20% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO2 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images..(BL3)		
Student Name:	Chavali Chandu	Reg. No.:	192472332
Section / Batch:		Date:	

Code of Conduct (192472332)
 I Chavali Chandu (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Chavali Chandu

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Grey Level Transformations	Enhancing image brightness and contrast using negative, log, power-law (gamma), and contrast stretching transformations.	1	10
Histogram Processing Techniques	Improving image contrast through histogram analysis, histogram equalization, and histogram matching.	2	10
Spatial Domain Enhancement	Reducing noise and enhancing image quality using smoothing filters and sharpening filters.	3	10
Frequency Domain Enhancement	Enhancing images by applying low-pass and high-pass filters.	4	10
Image Segmentation	Separating objects from the background using point, line, and edge detection, thresholding, region-based segmentation, edge linking, and boundary detection techniques.	5	10
GRAND TOTAL:		50 Marks	

Level Transformation

Purpose of Gray level transformation

Gray level transformation improves

visual quality of an image by changing pixel intensity values.

It is used to increase brightness, improve contrast, highlight important details, & prepare images for further processing.

1) Negative transformation

Formula:

$$s = L - 1 - r$$

Where:

L = number of gray levels.

r = Original pixel value

s = transformed pixel value.

Histogram Equalization

(a) Why is Histogram Equalization use

→ It improves image contrast by redistributing intensity values over the entire gray level range.

→ It enhances hidden details in low-contrast images.

Distribution Function

(b) Cumulative

(CDF)

Given histogram.

Intensity

frequency

0

2

3

6

10

6

total pixels = 26

Intensity

CDF

4

0

1

2

3

10

20

26

(c) Equalized using

$$S_k = \frac{CDF(k)}{24} \times (u-1)$$

Results:

→ 0 → 0

→ 1 → 1

→ 2 → 2

→ 3 → 3

Smoothing.

It removes noise while preserving as better.

It assigns higher weight to the center pixel & lower weight to neighboring pixels.

It produces smoother & more natural results.

Interpretation:

→ After applying the Gaussian kernel, the center pixel becomes the weighted average of neighboring pixels, reducing random noise & smoothing the image.

Edge Detection using Sobel operator

- Detects boundaries between objects.
- Identifies sudden changes in intensity.
- Used in image segmentation & object recognition.

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Interpretation:

- A large gradient value indicates a strong horizontal edge, while

a small gradient indicates a smooth region.

Contrast stretching.

- (a) Need for Contrast stretching.
- Improves visibility of image details.
 - Expands a narrow intensity range to the full gray-level range.
 - produces a clearer & sharper image.

(b) Formula

$$s = \frac{(r - 50) \times 255}{150 - 50}$$

where:

r = original pixel value.

s = new pixel value.

Interpretation:

→ pixel value originally between 50 & 150 are stretched to 0 - 255; significantly increasing image contrast.